THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 3: **The Atomic Foundations of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The molecular/formula unit mass of carbon dioxide, CO_2 is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 88 u **(b)** 46 u **(c)** 42 u **(d)** 44 u

2. The molecular/formula unit mass of sodium chloride, $NaCl$ (formula unit mass) is (H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40): 1

(a) 60.5 u **(b)** 117 u **(c)** 58.5 u **(d)** 56.5 u

3. 12 g of carbon burns in 32 g of oxygen. The mass of carbon dioxide formed is: 1

(a) 44 g **(b)** 20 g **(c)** 32 g **(d)** 12 g

4. In NaCl, Na : Cl is 23 : 35.5 by mass. 46 g of sodium needs how much chlorine? 1

(a) 46 g **(b)** 92 g **(c)** 35.5 g **(d)** 71 g

5. The chemical formula of sodium sulfate is: 1

(a) Na_2SO_4 **(b)** Na_2S **(c)** $NaSO_4$ **(d)** $Na(SO_4)_2$

6. The correct name of the covalent compound N_2O_4 is: 1

(a) dinitrogen tetraoxide **(b)** nitrogen oxide

(c) nitrogen tetroxide(II) **(d)** dinitrogen tetroxide

7. The Law of Constant (Definite) Proportions was proposed by: 1

(a) Dalton **(b)** Rutherford **(c)** Lavoisier **(d)** Joseph Proust

8. In an oxygen molecule O_2 the two atoms are joined by: 1

(a) a triple bond **(b)** an ionic bond **(c)** a single bond **(d)** a double bond

9. The bond between oppositely charged ions is called: 1

(a) covalent bond **(b)** hydrogen bond **(c)** ionic bond **(d)** metallic bond

10. Covalent compounds generally have: 1

(a) high electrical conductivity **(b)** no definite shape

(c) low melting and boiling points **(d)** high melting points

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. State the Law of Constant Proportions with an example. 2

12. 9 g of water is decomposed. What are the masses of hydrogen and oxygen obtained? 2

13. Why does the law of definite proportions apply to compounds and not to mixtures? 2

14. State any two postulates of Dalton's atomic theory. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Carbon monoxide has C : O = 3 : 4. How much oxygen combines with 9 g carbon? 3

16. Students X and Y made oxides of copper in the ratios Cu : O = 4 : 1 and 8 : 2. Do these results justify the law of constant proportions? 3

17. State the main postulates of Dalton's atomic theory (any four). 3

OR

How do Dalton's postulates explain 3
(i) the law of conservation of mass
(ii) the law of constant proportions?

18. Show the formation of a chlorine molecule by sharing of electrons. 3

OR

Show the formation of an oxygen molecule. What type of bond is formed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) State the postulates of Dalton's atomic theory. 5
(ii) How does it explain the laws of chemical combination?
(iii) Which postulate was later shown to be incorrect and why?

OR

(i) What is a covalent bond? 5
(ii) Show the formation of H_2, Cl_2 and HCl by sharing of electrons.
(iii) How many shared pairs are there in O_2 and N_2?

20. **Case study:** Water is formed when hydrogen and oxygen combine. 5
(i) Give the electronic configurations of H and O.
(ii) Explain the bonding in H_2O.
(iii) Why does water have different properties from H_2 and O_2?
(iv) What is the mass ratio of H : O in water?

OR

(i) Explain the formation of an ionic bond in NaCl. 5
(ii) Name the ions and write their electronic configurations.
(iii) Write the formula of magnesium oxide and explain how it forms.

END OF QUESTION PAPER — **Total: 40 Marks**

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